



Compliance & Secure Coding Practices in 25-26J-067

25-26J-067 Safe LAN

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1. Introduction

SafeLAN is a secure LAN-based file sharing and monitoring system designed to enhance confidentiality, integrity, and SafeLAN and accountability in file-sharing operations. The system integrates multiple advanced security components, including behavioral biometrics (keystroke dynamics and mouse movement monitoring), hand gesture-based authentication, face recognition, secure file transfer, and blockchain-based audit logging.

Given the sensitivity of user behavior data and file transactions, implementing compliance measures and secure coding practices is essential. These practices ensure that the system is resistant to vulnerabilities, protects user privacy, and maintains trustworthiness. Secure coding plays a critical role in handling biometric data, enforcing access control, and ensuring tamper-proof logging. Therefore, SafeLAN incorporates established standards and security techniques throughout its development lifecycle to provide a secure and reliable solution for LAN-based environments.

2. Secure Coding Standards

The SafeLAN system was developed by adhering to established secure coding standards to ensure the protection of sensitive data, secure authentication mechanisms, and reliable system behavior across all components. The project follows OWASP Secure Coding Practices to mitigate common vulnerabilities, including improper input validation, insecure authentication, and unauthorized access. These practices were applied not only to file handling and system access, but also to advanced components such as behavioral monitoring and biometric-based authentication. For example, secure input handling and validation were critical in processing keystroke dynamics and mouse movement data to prevent manipulation or injection of malicious inputs into the behavioral analysis modules.

In addition, ISO/IEC 27001 principles were considered to ensure proper information security management across the system, including secure storage, access control, and audit logging. Special attention was given to privacy-sensitive components such as face recognition and hand gesture-based authentication, ensuring that biometric data is securely processed and not unnecessarily exposed. Secure coding practices were also applied in continuous monitoring mechanisms, where real-time behavioral data is analyzed without compromising user privacy or system performance. Overall, these standards ensured that all modules of SafeLAN, from authentication to blockchain-based logging, operate securely, reliably, and in compliance with modern cybersecurity expectations.

3. Key Secure Coding Techniques Implemented

Input Validation: All inputs, including user credentials, file details, and behavioral data (keystrokes and mouse movements), are validated before processing. This prevents malicious data injections and ensures integrity in behavioral analysis and file operations.

Authentication & Authorization: SafeLAN uses multi-layered authentication:

- Behavioral biometrics (keystroke and mouse patterns) for continuous authentication
- Hand gesture-based verification for secure access
- Face recognition as an additional identity validation mechanism.
- Role-based access control ensures that only authorized users can access or modify files.

Encryption

- File data is protected using encryption techniques to ensure confidentiality.

- SHA-256 hashing is used in blockchain logging to maintain integrity.
- Cryptographic linking ensures logs cannot be altered once recorded.

Code Access Control

Access to system functionalities is restricted based on user roles and permissions. Unauthorized execution of file operations or system actions is prevented through strict access control mechanisms.

4. Tools & Technologies Used for Security

SafeLAN utilizes a combination of secure development practices and technologies to ensure system security. The application is developed as a secure desktop-based system with controlled interaction between modules such as authentication, behavioral monitoring, and blockchain logging. Cryptographic techniques such as SHA-256 hashing and digital signatures are used to ensure data integrity and authenticity within the system.

A permissioned blockchain network is used to enforce tamper-proof logging, ensuring that all file access events are securely recorded and cannot be modified. The system was tested within a controlled LAN environment to evaluate performance, security, and resistance to threats such as insider attacks and unauthorized access. Manual security testing and validation were performed to ensure proper implementation of authentication, encryption, and access control mechanisms across all components.

5. Compliance Standards

SafeLAN aligns with key information security and data protection requirements relevant to secure file systems and user monitoring. The system follows ISO/IEC 27001 principles to ensure secure access control, data protection, and auditability. Sensitive data, such as user behavior patterns and file access logs, is securely handled using encryption and controlled access mechanisms. Data protection is ensured by limiting the collection of user data to only what is necessary for authentication and monitoring. Biometric-related components such as face recognition and hand gesture authentication are implemented with consideration for user privacy, ensuring that data is not exposed or misused. Additionally, the blockchain-based logging system provides transparency and accountability by enabling tamper-proof audit trails, supporting compliance with security and auditing requirements. Proper documentation of system behavior and logging mechanisms further ensures that the project meets compliance expectations.

6. Challenges & Mitigation Strategies

One of the main challenges in SafeLAN was integrating multiple advanced security mechanisms, such as behavioral biometrics, gesture-based authentication, and blockchain logging, while maintaining system performance within a LAN environment. Ensuring secure handling of continuous behavioral data without affecting usability was also a significant challenge. Additionally, implementing tamper-proof logging using blockchain required careful design to avoid performance overhead.

These challenges were addressed by using a permissioned blockchain, which is optimized for LAN environments with low latency and controlled access. Lightweight cryptographic techniques and efficient data structures were used to ensure fast processing of logs and authentication data. Privacy concerns related to biometric data were mitigated by limiting data exposure and applying secure handling practices. Overall, a balance between security, performance, and usability was achieved through careful system design and adherence to secure coding principles.

7. References

- OWASP Secure Coding Practices – <https://owasp.org>
- CERT Secure Coding Standards – <https://www.cert.org/secure-coding/>
- ISO/IEC 27001 Information Security Standard
- GDPR Compliance Guidelines